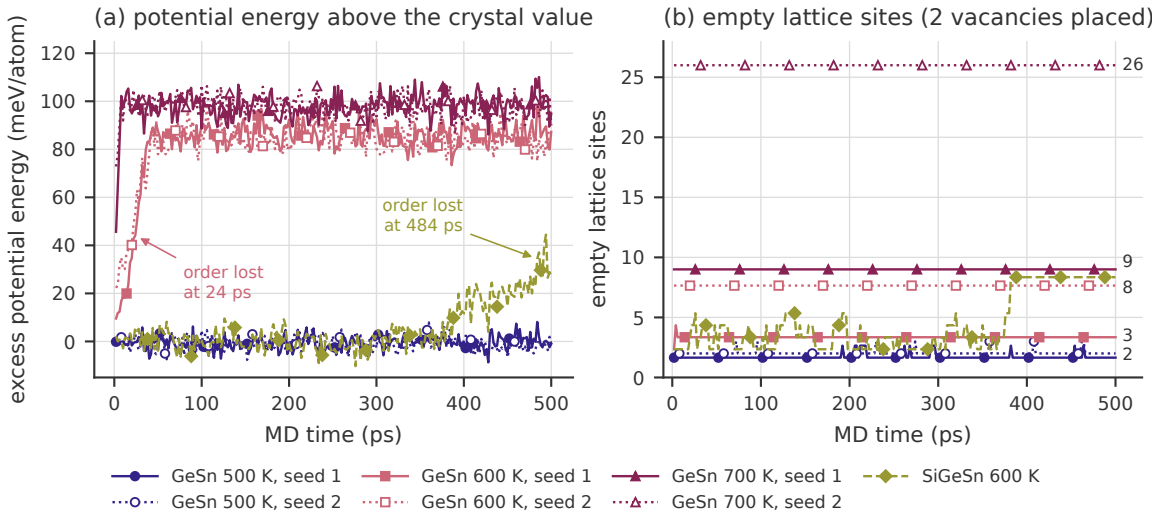
(a) potential energy above the crystal value
excess potential energy (meV/atom)
120
100
80
60
40
20
0
order lost
at 484 ps
order lost
at 24 ps
0
100
200
300
400
500
MD time (ps)
(b) empty lattice sites (2 vacancies placed)
empty lattice sites
25
20
15
10
5
0
26
9
8
3
2
0
100
200
300
400
500
MD time (ps)
GeSn 500 K, seed 1
GeSn 500 K, seed 2
GeSn 600 K, seed 1
GeSn 600 K, seed 2
GeSn 700 K, seed 1
GeSn 700 K, seed 2
SiGeSn 600 K